Safe Reinforcement learning for Peg-in-a-hole using Lagrangian SAC

Miao Xu, Ruining Wang
Lund University
Lund, Sweden

***Abstract*—Bimanual peg-in-hole assembly is a representative precision manipulation task that requires accurate coordination while maintaining safe clearance between the manipulators and the workspace. In conventional reward-only reinforcement learning, task performance and safety are typically combined within a single scalar reward, making policy behavior sensitive to the design and manual tuning of safety penalties. In this work, we formulate the task as a constrained Markov decision process that explicitly separates the task reward from a continuous clearance-based safety cost. We apply Lagrangian Soft Actor-Critic, which learns a dedicated cost critic and adaptively adjusts a Lagrange multiplier to account for constraint violations during policy optimization. The proposed method is evaluated against reward-only SAC in a simulated dual-arm manipulation environment with two seven-degree-of-freedom manipulators. Experimental results show that Lagrangian SAC preserves high task success while reducing cumulative safety cost, maximum constraint violations, and unsafe close-approach behavior. The constrained formulation also achieves competitive task returns without requiring a fixed manually tuned trade-off between task performance and safety. These results demonstrate the effectiveness of explicitly separating task objectives and safety constraints for safe bimanual precision manipulation.**

***Index Terms*—safe reinforcement learning, constrained reinforcement learning, bimanual manipulation, peg-in-hole assembly, Lagrangian SAC**

Supplementary Material

Code: `github.com/XuMiaoMiaoMiao/Safe_RL`

I. Introduction

Robotic assembly tasks require both precision and safety. Peg-in-hole is a representative precision manipulation problem demanding accurate relative positioning, alignment, and insertion under tight geometric tolerances. In bimanual manipulation, this becomes more challenging because two robot arms must coordinate in a shared workspace while moving close to each other and the environment, increasing the risk of arm-arm clearance violations, table contact, or collisions.

A common approach in standard reinforcement learning is to combine the task objective and safety penalties into a single scalar reward. Although simple to implement, this makes the trade-off between task performance and safety highly dependent on manual reward tuning. If the safety penalty is too small, the policy may frequently enter low-clearance regions; if too large, it may become overly conservative and fail to make sufficient task progress. Scalar reward shaping therefore does not provide a clean separation between what the robot should achieve and what constraints it must satisfy.

This project addresses the problem by formulating bimanual peg-in-hole manipulation as a Constrained Markov Decision Process (CMDP). The task reward and safety cost are defined separately: the reward drives geometric progress toward the insertion objective, while the cost measures constraint violations such as arm-arm and table-clearance violations. Hard physical contacts are treated as safety guards. This formulation allows task performance and safety behavior to be evaluated independently, rather than being conflated inside a single reward value.

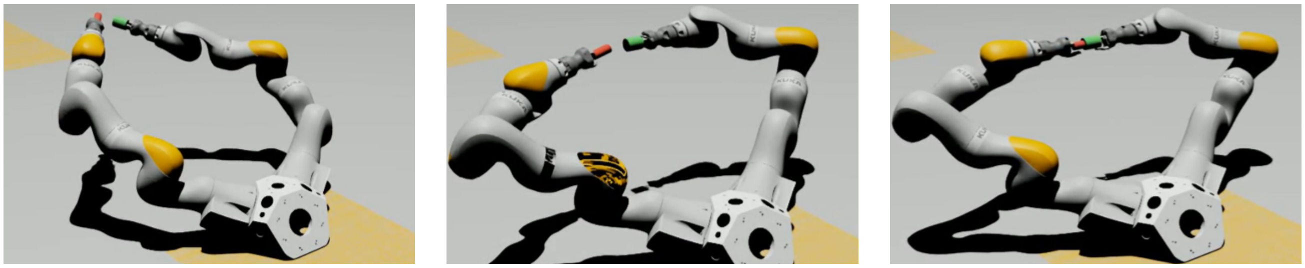
Based on this formulation, we compare standard Soft Actor-Critic (SAC) with a Lagrangian SAC variant. Both algorithms share the same task objective and safety guards, but differ in how safety costs are handled: standard SAC does not use the clearance cost during policy optimization; the cost is only logged post-hoc for evaluation and comparison, whereas Lagrangian SAC incorporates the clearance cost as a separate CMDP signal and uses an adaptive Lagrange multiplier to penalize constraint violations, providing a more principled mechanism for balancing task performance and safety.

The central question is whether constrained reinforcement learning can reduce clearance violations while maintaining task performance in bimanual peg-in-hole manipulation. The contributions are threefold: (1) we implement a bimanual peg-in-hole simulation task for studying safe RL; (2) we model clearance violation as a separate CMDP cost rather than a reward penalty; (3) we compare SAC and Lagrangian SAC under matched conditions to evaluate whether explicit constraint optimization can reduce safety violations while preserving task performance.

II. Related Work

Previous peg-in-hole studies have shown that deep reinforcement learning can solve high-precision assembly tasks under pose uncertainty [1]. Such tasks are widely used to study precision robotic manipulation because they require accurate relative motion, contact-aware behavior, and robustness to small pose errors. These properties make peg-in-hole a relevant benchmark for evaluating learning-based control methods in assembly scenarios.

Safe reinforcement learning is commonly formulated as a constrained Markov decision process (CMDP), where task return is optimized under cumulative cost constraints [2], [3]. This formulation separates the reward objective from the



Stage 1: prepositioning Stage 2: pre-axis alignment Stage 3: insertion

Fig. 1. Three-stage geometric curriculum for bimanual peg-in-hole manipulation. Stage 1 moves the peg tip to a pre-insertion pose in front of the hole. Stage 2 aligns the peg shaft with the hole axis. Stage 3 inserts the peg after sufficient radial and axial alignment.

safety signal and supports evaluation in terms of both performance and constraint satisfaction. Representative constrained RL methods include CPO, which optimizes policies under expected-cost constraints [3], IPO, which uses an interior-point formulation [4], and safe exploration benchmarks that report rewards and violations as separate metrics [5]. Other approaches address safety through action correction in continuous action spaces or long-term safety estimation under uncertainty [6], [7].

Soft Actor-Critic (SAC) is an off-policy maximum-entropy actor-critic algorithm that has been widely adopted for continuous-control problems [8]. Since standard SAC optimizes only a scalar reward objective, constrained SAC variants introduce additional cost or safety signals to explicitly account for constraint satisfaction. For example, WCSAC considers a worst-case safety objective for safety-constrained reinforcement learning [9]. Lagrangian methods provide another practical extension by incorporating an adaptive multiplier into the cost term, while PID Lagrangian methods further improve its responsiveness during training [10].

Building on these lines of work, we use MushroomRL [11] to compare SAC and Lagrangian SAC on a bimanual peg-in-hole safe-RL benchmark.

III. Problem Statement

We consider safe reinforcement learning for bimanual peg-in-hole manipulation with two 7-DoF KUKA iiwa manipulators. The left arm holds the peg, and the right arm holds the hole. The policy outputs a 14-dimensional continuous joint-velocity command,

$$a_t \in \mathbb{R}^{14}, \tag{1}$$

where the first seven dimensions control the peg arm and the remaining seven control the hole arm. Since both objects are actively controlled, the policy must coordinate the two manipulators and regulate the relative peg-hole geometry.

The task objective is defined by the relative pose between the peg and the hole. The peg tip must first reach a pre-insertion region, the peg shaft must align with the hole axis, and the peg must then advance into the hole without bad entry or penetration. The state s_t includes the joint positions and velocities of both arms, relative peg-hole geometric features, insertion-depth and radial-error measures, axis-alignment errors, and clearance-related quantities.

We formulate the problem as a constrained Markov decision process with state s_t, action a_t, task reward r_t, and safety cost c_t. The reward measures progress toward the peg-in-hole objective, whereas the cost measures violations of arm-arm and arm-table clearance margins. Proxy-based clearance violations generate continuous costs, while hard PhysX contacts are treated as absorbing safety events.

The constrained objective is

$$\max_{\pi} \mathbb{E}_{\pi}\left[\sum_{t=0}^{T-1} \gamma^t r_t\right], \quad \text{s.t.} \quad \mathbb{E}_{\pi}\left[\sum_{t=0}^{T-1} c_t\right] \le C_{\text{limit}}. \tag{2}$$

This formulation separates task performance from safety constraint satisfaction.

IV. Approach

We compare reward-only SAC and Lagrangian SAC under a matched environment, reward function, and hard-contact guard. Both methods optimize the same bimanual peg-in-hole task reward. The key difference is that SAC does not incorporate the clearance cost during policy optimization; the cost is only logged post-hoc for evaluation and comparison. By contrast, Lagrangian SAC optimizes a CMDP cost during training through an adaptive Lagrange multiplier.

A. Geometric Curriculum and Clearance Cost

The task is trained with a three-stage geometric curriculum that follows the physical ordering of bimanual peg-in-hole assembly, as illustrated in Fig. 1. Stage 1 learns prepositioning, where the peg tip is moved to a pre-insertion pose in front of the hole. Stage 2 adds shaft-level centering and pre-axis alignment, making the peg pose geometrically compatible with insertion. Stage 3 activates insertion-specific objectives, where the peg is advanced into the hole only after sufficient radial and axial alignment. These stages are trained sequentially, with each stage warm-started from the converged policy of the previous one; full training details are provided in Appendix B.

This curriculum first establishes a useful relative peg-hole pose, then refines the shaft and axis geometry, and finally rewards insertion progress under clean alignment conditions, thereby discouraging policies that increase insertion depth under poor radial or axial alignment.

For curriculum stage k, the task reward is written as

$$r_t^{(k)} = \sum_{i \in \mathcal{A}_k} w_i r_i(t) + r_{\text{reg}}(t), \tag{3}$$

TABLE I
ACTIVE REWARD COMPONENTS IN THE GEOMETRIC CURRICULUM.

Component	Objective	Stage 1	Stage 2	Stage 3
r_d	Depth tracking	✓	✓	✓
r_{tip}	Tip radial centering	✓	✓	–
$r_{\max}$	Shaft radial centering	–	✓	✓
r_{axis}	Axis alignment	–	✓	✓
$r_{\Delta\phi}$	Gated insertion progress	–	–	✓
r_{soft}	Clean-insertion shaping	–	–	✓
r_{pen}	Penetration penalty	–	–	✓
r_{bad}	Bad-entry penalty	–	–	✓
r_{reg}	Motion regularization	✓	✓	✓

where $\mathcal{A}_k$ is the set of active reward components, w_i is the corresponding weight, and r_{reg} is a motion regularization term. The active reward components are summarized in Table I.

The clearance cost is independent of the task reward. Let h_t^{arm} be the minimum clearance between the two arm proxy geometries, and let h_t^{table} be the minimum clearance between the robot proxies and the table. Given clearance margins m_{arm} and m_{table}, the per-step cost is

$$c_t = \max\left(\left[\frac{m_{\text{arm}} - h_t^{\text{arm}}}{m_{\text{arm}}}\right]_+, \left[\frac{m_{\text{table}} - h_t^{\text{table}}}{m_{\text{table}}}\right]_+\right), \quad (4)$$

where $[x]_+ = \max(x, 0)$. The cost is zero when all clearances exceed their margins and becomes positive when any margin is violated. It is not added to the task reward and does not terminate the episode. Hard PhysX contacts are handled separately as absorbing safety events.

Thus, the environment provides three distinct signals: a task reward for assembly progress, a continuous clearance cost for pre-contact safety, and hard-contact guards for collisions.

B. SAC Baseline and Lagrangian SAC

The SAC baseline optimizes the maximum-entropy task objective using only the scalar task reward. It does not use the clearance cost during training; safety metrics are recorded only for evaluation.

Lagrangian SAC augments SAC with a cost critic Q_C and a nonnegative Lagrange multiplier λ. The replay buffer stores transitions of the form

$$(s_t, a_t, r_t, c_t, s_{t+1}, \delta_t), \quad (5)$$

where δ_t is the termination flag. The reward critic Q_R estimates task return, and the cost critic Q_C estimates cumulative clearance cost.

The actor objective is

$$\mathcal{L}_\pi = \mathop{\mathbb{E}}_{\substack{s_t \sim \mathcal{D} \\ a_t \sim \pi}} \left[\alpha \log \pi(a_t|s_t) - Q_R(s_t, a_t) + \lambda Q_C(s_t, a_t)\right] \quad (6)$$

The term $-Q_R$ encourages task completion, while λQ_C penalizes actions with high predicted clearance cost. The multiplier is updated from completed rollout episodes as

$$\log \lambda \leftarrow \log \lambda + \eta_\lambda \left(\bar{C}_{\text{episode}} - C_{\text{limit}}\right), \quad (7)$$

where $\bar{C}_{\text{episode}}$ is the observed episode cost. When the episode cost exceeds the limit, λ increases and the policy places greater weight on safety. When the cost remains below the limit, λ decreases and the policy emphasizes task progress.

This adaptive update replaces a fixed manually tuned safety penalty with an online constraint-balancing mechanism.

V. EXPERIMENTS AND RESULTS

We compare reward-only SAC and Lagrangian SAC on the proposed bimanual peg-in-hole benchmark. The simulation environment is built in NVIDIA Isaac Sim, and the SAC-based training pipeline is implemented in MushroomRL [11].

Both methods use the same observation space, action space, task reward, hard-contact guards, and three-stage curriculum. They differ only in whether the clearance cost is optimized: SAC records it only for evaluation, whereas Lagrangian SAC uses it as a CMDP cost.

Following safe-RL evaluation practice, we report task performance using discounted return and success rate, and safety using episodic constraint cost and maximum violation [7].

A. Full Curriculum Rollout

We first verify whether the three-stage curriculum can support the complete bimanual peg-in-hole sequence. As shown in Fig. 2, both SAC and Lagrangian SAC reach near-perfect success in Stage 1 prepositioning, Stage 2 pre-axis alignment, and Stage 3 insertion.

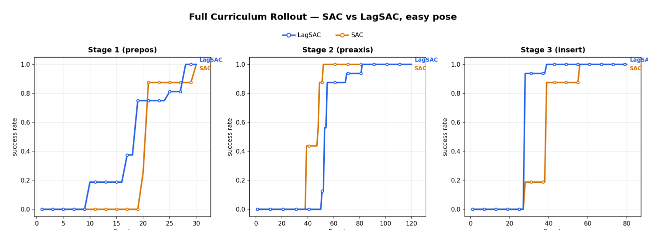


Fig. 2. Full-curriculum rollout on the easy pose. Success is evaluated using the active geometric success mask of each curriculum stage.

The success rate is computed using the active stage-specific geometric condition. Stage 1 checks pre-insertion depth and radial accuracy, Stage 2 additionally checks shaft centering and axis alignment, and Stage 3 further checks insertion depth, bounded depth error, alignment, and penetration. A rollout is counted as successful only when the active condition is held for the required number of consecutive steps.

This rollout is used as a task-feasibility check rather than the main safety benchmark. Since most low-clearance motions occur when the arms first enter the shared workspace, the controlled safety comparison is conducted on Stage 1.

B. Stage 1 Initial-Pose Benchmark

The main benchmark evaluates Stage 1 pre-positioning under two reset conditions.

The easy pose tests whether the constrained formulation preserves task performance under regular initialization.

The harder pose increases the likelihood of close-approach motions and is used as the primary safety comparison.

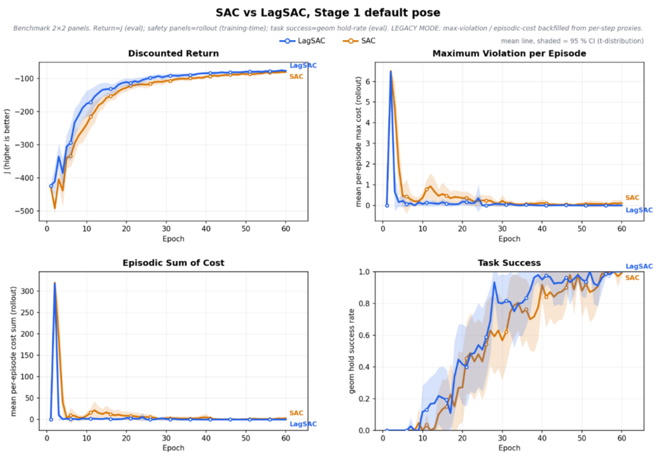


Fig. 3. Stage 1 easy-pose benchmark curves for SAC and Lagrangian SAC.

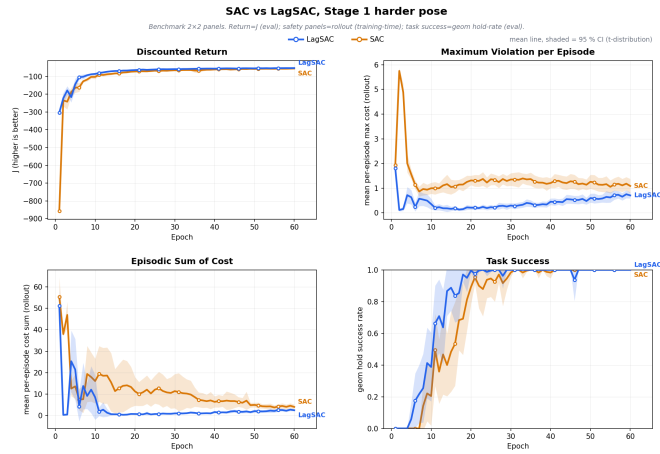


Fig. 4. Stage 1 harder-pose benchmark curves for SAC and Lagrangian SAC.

TABLE II
INITIAL-POSE SAFETY BENCHMARK RESULTS. ALL VALUES ARE REPORTED AS MEAN WITH 95% CONFIDENCE INTERVALS OVER 15 SEEDS.

Initial pose	Metric	SAC	Lagrangian SAC
Easy	Return J	-79.7 [-82.1, -77.3]	-76.5 [-78.7, -74.3]
	Success rate	0.992 [0.978, 1.000]	1.000 [1.000, 1.000]
	Episodic constraint cost	2.80 [0.00, 6.52]	0.025 [0.000, 0.067]
	Maximum violation	0.113 [0.000, 0.242]	0.003 [0.000, 0.008]
Harder	Return J	-55.61 [-56.72, -54.50]	-52.85 [-53.70, -52.01]
	Success rate	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
	Episodic constraint cost	4.011 [2.963, 5.059]	2.456 [2.043, 2.869]
	Maximum violation	1.090 [0.827, 1.353]	0.708 [0.595, 0.821]

In the easy pose, both methods solve the pre-positioning task, as shown in Fig. 3 and Table II. SAC reaches a success rate of 0.992, while Lagrangian SAC reaches 1.000.

The main difference appears in safety. Lagrangian SAC reduces the episodic constraint cost from 2.80 to 0.025 and the maximum violation from 0.113 to 0.003, indicating that it removes most residual low-clearance behavior without reducing task performance.

The harder pose provides the main safety comparison, as shown in Fig. 4 and Table II. Both algorithms reach full Stage 1 success, but this initialization leads to larger clearance violations.

Lagrangian SAC still reduces the episodic constraint cost from 4.011 to 2.456 and the maximum violation from 1.090 to 0.708, while achieving a higher return, from -55.61 to -52.85.

Overall, both initial poses show the same trend: Lagrangian SAC preserves task success while reducing clearance-related safety violations. The reduction is nearly complete in the easy pose and remains clear in the harder pose, where close-approach motions are more likely.

VI. CONCLUSION

To address the limitation of reward-only reinforcement learning, where task progress and safety penalties are combined into a single scalar reward, we formulate the insertion task as a Constrained Markov Decision Process (CMDP) that separates the task reward from the safety cost. Specifically, Lagrangian SAC extends SAC with a cost critic and a Lagrangian multiplier, allowing constraint violations to be optimized through an adaptive penalty rather than a fixed, manually tuned reward term.

The experimental results show that, under the same environment, task reward, and hard safety guards, Lagrangian SAC achieves better safety while maintaining task performance comparable to the SAC baseline. It preserves the task success rate while reducing the episodic clearance cost and the maximum constraint violation. These results indicate that explicitly modeling safety costs is more effective than relying solely on reward shaping in this bimanual assembly task.

Future work should extend the current constrained peg-in-hole benchmark toward more dynamic manipulation tasks, compare with broader safe RL and safety-filter methods, and introduce torque-level constraints for safer contact-rich manipulation.

REFERENCES

[1] T. Inoue, G. D. Magistris, A. Munawar, T. Yokoya, and R. Tachibana, "Deep reinforcement learning for high precision assembly tasks," in *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017, pp. 819–825. [Online]. Available: https://research.ibm.com/publications/deep-reinforcement-learning-for-high-precision-assembly-tasks

[2] E. Altman, "Constrained markov decision processes with total cost criteria: Lagrangian approach and dual linear program," *Mathematical Methods of Operations Research*, vol. 48, no. 3, pp. 387–417, 1998.

[3] J. Achiam, D. Held, A. Tamar, and P. Abbeel, “Constrained policy optimization,” in *Proceedings of the 34th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 70. PMLR, 2017, pp. 22–31. [Online]. Available: https://proceedings.mlr.press/v70/achiam17a.html

[4] Y. Liu, J. Ding, and X. Liu, “IPO: Interior-point policy optimization under constraints,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 34, no. 04, 2020, pp. 4940–4947.

[5] A. Ray, J. Achiam, and D. Amodei, “Benchmarking safe exploration in deep reinforcement learning,” 2019. [Online]. Available: https://arxiv.org/abs/1910.01708

[6] G. Dalal, K. Dvijotham, M. Vecerik, T. Hester, C. Paduraru, and Y. Tassa, “Safe exploration in continuous action spaces,” 2018. [Online]. Available: https://arxiv.org/abs/1801.08757

[7] J. Günster, P. Liu, J. Peters, and D. Tateo, “Handling long-term safety and uncertainty in safe reinforcement learning,” in *Proceedings of The 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2025, pp. 4670–4697. [Online]. Available: https://proceedings.mlr.press/v270/gunster25a.html

[8] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, “Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor,” in *Proceedings of the 35th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 80. PMLR, 2018, pp. 1861–1870. [Online]. Available: https://arxiv.org/abs/1801.01290

[9] Q. Yang, T. D. Simao, S. H. Tindemans, and M. T. J. Spaan, “WCSAC: Worst-case soft actor critic for safety-constrained reinforcement learning,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, no. 12, 2021, pp. 10 639–10 646. [Online]. Available: https://ojs.aaai.org/index.php/AAAI/article/view/17272

[10] A. Stooke, J. Achiam, and P. Abbeel, “Responsive safety in reinforcement learning by PID lagrangian methods,” in *Proceedings of the 37th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 119. PMLR, 2020, pp. 9133–9143. [Online]. Available: https://proceedings.mlr.press/v119/stooke20a.html

[11] C. D’Eramo, D. Tateo, A. Bonarini, M. Restelli, and J. Peters, “MushroomRL: Simplifying reinforcement learning research,” *Journal of Machine Learning Research*, vol. 22, no. 131, pp. 1–5, 2021. [Online]. Available: https://www.jmlr.org/papers/v22/18-056.html

APPENDIX A
ALGORITHMS

Algorithms 1 and 2 summarize the SAC baseline and Lagrangian SAC used for comparison in this work.

Algorithm 1 SAC baseline for bimanual peg-in-hole

Initialize: policy π_θ, reward critics $Q_{\omega_1}, Q_{\omega_2}$, target critics $Q_{\bar{\omega}_1}, Q_{\bar{\omega}_2}$, replay buffer $\mathcal{D}$, temperature α.
Input: environment $\mathcal{E}$, epochs N, discount factor γ, soft-update rate τ.

for $e = 1, \ldots, N$ **do**
  Collect transitions with $a_t \sim \pi_\theta(\cdot|s_t)$ and observe $(s_{t+1}, r_t, d_t, \text{info}_t)$.
  Store $(s_t, a_t, r_t, s_{t+1}, d_t)$ in $\mathcal{D}$.
  Log $\text{info}_t[\text{cost}]$ only for safety diagnostics.
  if $\mathcal{D}$ is ready **then**
    Sample mini-batch $(s, a, r, s', d) \sim \mathcal{D}$.
    Update reward critics with the SAC Bellman target:

$$y_R = r + \gamma(1-d)\left[\min_i Q_{\bar{\omega}_i}(s', a') - \alpha \log \pi_\theta(a'|s')\right].$$

    Update the actor by minimizing:

$$\mathcal{L}_\pi = \alpha \log \pi_\theta(\tilde{a}|s) - \min_i Q_{\omega_i}(s, \tilde{a}), \quad \tilde{a} \sim \pi_\theta(\cdot|s).$$

    Update temperature α and soft-update target critics: $\bar{\omega}_i \leftarrow \tau\omega_i + (1-\tau)\bar{\omega}_i$.
  end if
  Evaluate deterministic policy and log task and safety metrics.
end for

Algorithm 2 Lagrangian SAC with clearance-cost constraint

Initialize: policy π_θ, reward critics $Q^R_{\omega_1}, Q^R_{\omega_2}$, cost critics $Q^C_{\psi_1}, Q^C_{\psi_2}$, target critics, replay buffer $\mathcal{D}$, temperature α, Lagrange multiplier λ.
Input: CMDP environment $\mathcal{E}$, epochs N, cost limit $\bar{C}$, multiplier learning rate η_λ.

for $e = 1, \ldots, N$ **do**
  Reset the rollout cost tracker.
  Collect transition with $a_t \sim \pi_\theta(\cdot|s_t)$ and observe $(s_{t+1}, r_t, d_t, \text{info}_t)$.
  Read cost $c_t = \text{info}_t[\text{cost}]$ and store $(s_t, a_t, r_t, c_t, s_{t+1}, d_t)$ in $\mathcal{D}$.
  Accumulate episodic cost $\sum_t c_t$ and maximum violation $\max_t c_t$.
  if $\mathcal{D}$ is ready **then**
    Sample mini-batch $(s, a, r, c, s', d) \sim \mathcal{D}$.
    Update reward and cost critics using

$$y_R = r + \gamma(1-d)\left[\min_i Q^R_{\bar{\omega}_i}(s', a') - \alpha \log \pi_\theta(a'|s')\right],$$

$$y_C = c + \gamma_C(1-d)\max_i Q^C_{\bar{\psi}_i}(s', a').$$

    Update the actor with the Lagrangian SAC objective:

$$\mathcal{L}_\pi = \alpha \log \pi_\theta(\tilde{a}|s) - \min_i Q^R_{\omega_i}(s, \tilde{a}) + \lambda \max_i Q^C_{\psi_i}(s, \tilde{a}),$$

    Update temperature α and soft-update reward and cost target critics.
  end if
  if rollout episodes have completed **then**
    Compute mean episodic cost $\hat{C}_e$.
    Update the multiplier:

$$\log \lambda \leftarrow \text{clip}\left(\log \lambda + \eta_\lambda(\hat{C}_e - \bar{C}), \log \lambda_{\min}, \log \lambda_{\max}\right).$$

  end if
  Evaluate deterministic policy and log task, safety, and λ metrics.
end for

APPENDIX B
BIMANUAL PEG-IN-HOLE ENVIRONMENT

This appendix provides the environment-level reward and cost specification. All costs are continuous; a positive cost indicates the magnitude of a clearance-margin violation. All reward terms below contribute only to the task reward and are not used as safety costs.

A. Task

The environment consists of two 7-DoF manipulators. The left arm holds the peg, and the right arm holds the hole. The objective is to coordinate both arms such that the peg tip first reaches a pre-insertion pose, the peg shaft aligns with the hole axis, and the peg is then inserted into the hole without bad entry or penetration.

The task is trained with three curriculum stages: Stage 1 trains prepositioning, Stage 2 trains pre-axis alignment, and Stage 3 trains insertion. These stages are optimized sequentially rather than as independent runs. Each stage is optimized to its own stage-specific geometric success criterion, and its converged policy is used to warm-start the next stage. Specifically, Stage 1 is trained from scratch, Stage 2 initializes from the Stage 1 policy, and Stage 3 initializes from the Stage 2 policy. This policy transfer mechanism reuses the prepositioning and alignment skills already acquired, allowing each stage to learn only the incremental geometry it introduces. Throughout all stages, the same environment and reward definitions are used for both SAC and Lagrangian SAC, enabling direct algorithm comparison.

B. State and Action Space

The policy outputs a 14-dimensional continuous joint-velocity command,

$$a_t \in \mathbb{R}^{14}, \tag{8}$$

where the first seven dimensions control the peg arm and the remaining seven dimensions control the hole arm. The observation includes the joint positions and velocities of both arms, the relative peg-hole geometry, the insertion-depth coordinate, radial and axis-alignment errors, and clearance-related quantities.

C. Reward

Let p_t^{tip} be the peg-tip position, p_t^{entry} the hole-entry position, and z_t^{hole} the outward direction of the hole axis. The signed depth of the peg tip is

$$d_t = -\left(p_t^{\text{tip}} - p_t^{\text{entry}}\right)^\top z_t^{\text{hole}}. \tag{9}$$

Thus, $d_t < 0$ indicates that the peg tip is in front of the hole entrance, and $d_t > 0$ indicates that it has entered the hole region.

Let ρ_t^{tip} be the radial distance from the peg tip to the hole axis. Let $\rho_t^{\max}$ be the maximum radial distance over sampled points on the peg shaft. The first quantity is used for tip-level prepositioning, while the second measures shaft-level centering.

Let z_t^{peg} be the peg-axis direction. Since the desired peg and hole axes are anti-parallel, the axis-alignment error is

$$e_t^{\text{axis}} = 1 + \left(z_t^{\text{peg}}\right)^\top z_t^{\text{hole}}. \tag{10}$$

Perfect alignment gives $e_t^{\text{axis}} = 0$. Let p_t^{pen} denote the geometric penetration between the peg and the hole wall.

The depth-tracking term is

$$r_d(t) = -\operatorname{clip}\left(|d_t - d_t^\star|, 0, d_{\text{sat}}\right), \tag{11}$$

where $d_t^\star$ is the stage-dependent target depth, and d_{sat} bounds the depth penalty.

The tip-centering and shaft-centering terms are

$$r_{\text{tip}}(t) = -\operatorname{clip}\left(\rho_t^{\text{tip}}, 0, \rho_{\text{sat}}\right), \tag{12}$$

$$r_{\max}(t) = -\operatorname{clip}\left(\rho_t^{\max}, 0, \rho_{\text{sat}}\right), \tag{13}$$

where ρ_{sat} bounds the radial penalty.

The axis-alignment and penetration terms are

$$r_{\text{axis}}(t) = -e_t^{\text{axis}}, \qquad r_{\text{pen}}(t) = -p_t^{\text{pen}}. \tag{14}$$

Insertion progress is gated by the current alignment quality. The alignment gate is

$$G_t = \exp\left(-\left(\frac{\rho_t^{\max}}{\sigma_\rho}\right)^2 - \left(\frac{e_t^{\text{axis}}}{\sigma_a}\right)^2 - \left(\frac{p_t^{\text{pen}}}{\sigma_p}\right)^2\right), \tag{15}$$

where σ_ρ, σ_a, and σ_p set the radial, axis, and penetration scales. The normalized gated insertion progress is

$$\phi_t = G_t \operatorname{clip}\left(\frac{d_t - d_{\min}}{d_{\max} - d_{\min}}, 0, 1\right), \tag{16}$$

where $d_{\min}$ and $d_{\max}$ define the insertion-progress interval. The incremental insertion-progress reward is

$$r_{\Delta\phi}(t) = \phi_t - \phi_{t-1}. \tag{17}$$

This term rewards forward insertion only when the peg is sufficiently aligned.

The clean-insertion shaping term is

$$\begin{aligned} r_{\text{soft}}(t) = \exp\Bigg(&-\left(\frac{|d_t - d_t^\star|}{\sigma_d}\right)^2 - \left(\frac{\rho_t^{\max}}{\sigma_{\rho,s}}\right)^2 \\ &- \left(\frac{e_t^{\text{axis}}}{\sigma_{a,s}}\right)^2 - \left(\frac{p_t^{\text{pen}}}{\sigma_{p,s}}\right)^2\Bigg), \end{aligned} \tag{18}$$

where σ_d, $\sigma_{\rho,s}$, $\sigma_{a,s}$, and $\sigma_{p,s}$ determine the width of the local shaping well. This term gives a positive reward near clean insertion configurations.

The bad-entry penalty discourages insertion under poor geometry. The depth activation is

$$\eta_d(t) = \operatorname{clip}\left(\frac{d_t}{d_{\text{ins}}}, 0, \eta_{\max}\right), \tag{19}$$

where d_{ins} controls how the penalty is activated after entering the hole region and $\eta_{\max}$ upper-bounds the activation. The radial, axis, and penetration violation terms are

$$v_\rho(t) = \left[\frac{\rho_t^{\max}}{\rho_{\text{safe}}} - 1\right]_+, \tag{20}$$

$$v_a(t) = \left[\frac{e_t^{\text{axis}}}{e_{\text{safe}}} - 1\right]_+, \tag{21}$$

$$v_p(t) = \left[\frac{p_t^{\text{pen}}}{p_{\text{safe}}} - 1\right]_+. \tag{22}$$

Here, ρ_{safe}, e_{safe}, and p_{safe} are the radial, axis, and penetration thresholds. The bad-entry penalty is

$$r_{\text{bad}}(t) = -\eta_d(t)\operatorname{clip}\left(v_\rho(t) + v_a(t) + v_p(t), 0, v_{\max}\right), \tag{23}$$

where $v_{\max}$ limits the penalty magnitude. This penalty is active only when the peg has entered the hole region and at least one geometric error exceeds its threshold.

The regularization term is

$$r_{\text{reg}}(t) = -w_a\|a_t\|_2^2 - w_h H(q_t) - w_j J(q_t), \tag{24}$$

where q_t is the dual-arm joint configuration. $H(q_t)$ penalizes deviation from a nominal home posture, and $J(q_t)$ penalizes proximity to joint limits.

At curriculum stage k, the task reward is

$$r_t^{(k)} = \sum_{i \in \mathcal{A}_k} w_i r_i(t) + r_{\text{reg}}(t), \tag{25}$$

where the active set $\mathcal{A}_k$ is given in Table I.

D. Cost and Termination

The clearance cost is computed from proxy distances. Let h_t^{arm} be the minimum clearance between the two arm proxy geometries, and let h_t^{table} be the minimum clearance between the robot proxies and the table. Given clearance margins m_{arm} and m_{table}, the cost is

$$c_t = \max\left(\left[\frac{m_{\text{arm}} - h_t^{\text{arm}}}{m_{\text{arm}}}\right]_+, \left[\frac{m_{\text{table}} - h_t^{\text{table}}}{m_{\text{table}}}\right]_+\right). \tag{26}$$

A positive value indicates a violation of at least one clearance margin. This cost is continuous, independent of all reward terms, and does not terminate the episode.

Hard PhysX arm or table contacts are treated as absorbing safety events. Success indicators are not added as hard reward terms; they are used only for evaluation, logging, and checkpoint selection.